The fact is that if you thought about anatomy, physiology, or almost any aspect of biology, not just behaviour, in Sahlins's way you would arrive at his same non-existent problem. The embryological development of any bit of an animal's or plant's body requires complicated mathematics for its complete description, but this does not mean that the animal or plant must itself be a clever mathematician! Very tall trees usually have huge buttresses flaring out like wings from the base of their trunks. Within any one species, the taller the tree, the relatively larger the buttresses. It is widely accepted that the shape and size of these buttresses are close to the economic optimum for keeping the tree erect, although an engineer would require quite sophisticated mathematics to demonstrate this. It would never occur to Sahlins or anyone else to doubt the theory underlying buttresses simply on the grounds that trees lack the mathematical expertise to do the calculations. Why, then, raise the problem for the special case of kin-selected behaviour? It can't be because it is behaviour as opposed to anatomy, because there are plenty of other examples of behaviour (other than kin-selected behaviour, I mean) that Sahlins would cheerfully accept without raising his 'epistemological' objection; think, for instance, of my own illustration (pp. 124–5) of the complicated calculations that in some sense we all must do whenever we catch a ball. One cannot help wondering: are there social scientists who are quite happy with the theory of natural selection generally but who, for quite extraneous reasons that may have roots in the history of their subject, desperately want to find something—*anything*—wrong with the theory of *kin selection specifically*?

p. 128* … *we have to think how animals might actually go about estimating who their close relations are … We know who our relations are because we are told …

The whole subject of kin recognition has taken off in a big way since this book was written. Animals, including ourselves, seem to show remarkably subtle abilities to discriminate relatives from nonrelatives, often by smell. A recent book, *Kin Recognition in Animals*, summarizes what is now known. The chapter on humans by Pamela Wells shows that the statement above ('*We* know who our relations are because we are told') needs to be supplemented: there is at least circumstantial evidence that we are capable of using various nonverbal cues, including the smell of our relatives' sweat. The whole subject is, for me, epitomized by the quotation with which she begins:

> all good kumrads you can tell
> by their altruistic smell
> e. e. cummings

Relatives might need to recognize one another for reasons other than altruism. They might also want to strike a balance between outbreeding and inbreeding, as we shall see in the next note.

p. 129* … *the injurious effects of recessive genes which appear with inbreeding. (For some reason many anthropologists do not like this explanation.)

A lethal gene is one that kills its possessor. A recessive lethal, like any recessive gene, doesn't exert its effect unless it is in double dose. Recessive lethals get by in the gene pool, because most individuals possessing them have only one copy and therefore never suffer the effects. Any given lethal is rare, because if it ever gets common it meets copies of itself and kills off its carriers. There could nevertheless be lots of different types of lethal, so we could still all be riddled with them. Estimates vary as to how many different ones there are lurking in the human gene pool. Some books reckon as many as two lethals, on average, per person. If a random male mates with a random female, the chances are that his lethals will not match hers and their children will not suffer. But if a brother mates with a sister, or a father with a daughter, things are ominously different. However rare my lethal recessives may be in the population at large, and however rare